

The Parasitic Threshold and Boundary Logic: Defense Pressure Before Modern Immunity

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Abstract

Parasitism becomes meaningful only after replication, retention, and container inheritance exist. This manuscript therefore moves the parasitic threshold after the protocell integration gate (Paper 9). In CDFD terms, a parasitic sequence or pathway is an internal flux drain that consumes activated resources while failing to maintain the host network or its boundary. The key claim is conditional and testable: once copying and compartment persistence are coupled, exploitative variants create selection pressure for specificity, exclusion, proofreading, degradation, and boundary control. The paper does not claim literal modern viruses, immune systems, or oncology at the dawn of life — it identifies the earliest functional problem that later defense systems solve.

Keywords: parasitic threshold; boundary logic; prebiotic defense; replication; origins of life.

Preprint note

This is a computational research article. The companion script produces numerical diagnostics reported in the Computational Results section. It is not a narrative review and does not claim historical proof of a unique abiogenesis pathway.

Symbol Conventions

See Paper 1, §Symbol Conventions, for the base CDFD and Greek-symbol tables. This paper introduces Φ_{host} , Φ_{loss} , and the host-survival inequality (§4).

1 Introduction

Prior protocell-integration work established the minimum machinery required for protocellular evolution: information must copy with bounded error, and container success must covary with

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internal chemistry. Once that machinery exists, exploitation becomes unavoidable in principle. Any variant that uses the replication or catalytic apparatus without contributing to persistence can grow faster than cooperative chemistry over short horizons.

This is the parasitic threshold. It is not a statement that modern viruses existed before cells. It is a statement about a failure mode of coupled autocatalytic and replicative systems, adjacent to classic error-threshold and major-transition logic [1, 2].

2 Parasitic Flux Drain

Let the cooperative autocatalytic/replicative network have operational flux Φ_{host} . A parasitic component consumes resources at rate Φ_{loss} while contributing little or nothing to retention, repair, catalytic closure, or boundary persistence. The effective host flux becomes

$$\Phi_{\text{eff}} = \Phi_{\text{host}} - \Phi_{\text{loss}}. \quad (1)$$

The adaptive ratio becomes

$$\Psi_s = \left(\frac{\Phi_{\text{eff}}}{C_{\text{mem}}} \right) SM_s. \quad (2)$$

If Φ_{loss} grows faster than host repair or exclusion, the system crosses from productive replication to starvation, washout, or loss of inherited organization.

3 Boundary Logic

The response cannot be only “more flux.” Extra substrate can amplify the parasite as easily as the host. A surviving protocell-like system needs constraint changes:

- selective permeability at the boundary,
- catalytic specificity inside the compartment,
- degradation or sequestration of noncontributing templates,
- copying rules that preserve useful heredity while limiting runaway freeloaders.

These are early functional analogues of defense logic. They are not yet modern immune systems.

4 Mathematical Gate

A simple host-survival inequality is

$$\Gamma_{\text{host}} Q_{\text{host}} R_{\text{ret}} > \Gamma_{\text{parasite}} Q_{\text{parasite}} R_{\text{access}}, \quad (3)$$

where R_{ret} measures host retention in the container and R_{access} measures parasite access to host resources. Boundary evolution acts by lowering R_{access} or increasing the cost of parasitic replication.

The remaining symbols are operational. Γ_{host} is the productive host replication or catalytic-throughput rate, Q_{host} is host copying fidelity or functional retention per copy, Γ_{parasite} is the parasite replication or drain rate, and Q_{parasite} is parasite persistence per copy. The inequality is not a calibrated law; it is a gate that says cooperative persistence must outpace exploitative access after boundary effects are included.

The falsifiable implication is direct: in a model or experiment with heritable compartments, increasing noncontributing replicators should collapse cooperative persistence unless some specificity or exclusion mechanism is introduced.

5 Validation Design

The experimental target is a compartmental copying system in which host-like templates or catalysts contribute to persistence while parasite-like variants consume the same activated resources without contributing to boundary survival. The controls should hold total template concentration, activated substrate, compartment size distribution, and environmental cycling fixed while varying only parasite fraction and boundary specificity.

Useful observables include host-template fraction after serial transfer, compartment survival, product yield per compartment, leakage, and the covariance between internal chemistry and boundary growth or division. The CDFD expectation is a threshold response: increasing parasite access should sharply reduce host persistence unless selective permeability, degradation, sequestration, or copying specificity is present. Failure to see such a threshold under otherwise heritable conditions would weaken this paper’s claim.

6 Computational Results: Spatial Boundary Gate

The final release expands the companion script beyond a well-mixed host–parasite ODE. A second diagnostic places host-like and parasite-like fields on a two-dimensional compartment grid. The boundary gate modifies parasite access to the interior while the top of the domain receives resource inflow. This remains a toy stress test, but it makes the physical point visible: boundary specificity is not just a scalar penalty; it changes where resource-consuming variants can spatially persist.

The two-dimensional panel should be read as a falsification target. If parasite suppression appears only in a well-mixed equation but disappears once diffusion, leakage, and boundary access are represented spatially, the boundary-logic claim is too weak. Conversely, if gated compartments retain host-dominant interiors across parameter sweeps while open boundaries do not, the model earns a stronger experimental target.

7 Conclusion

The parasitic threshold is the first crisis created by successful protocell integration. Once copying, retention, and compartment inheritance exist, exploitative variants force the system to evolve boundary logic. Photochemical expansion is treated as a later energy-input channel (Paper 11) rather than a prerequisite for the parasitic-threshold argument.

8 Reproducibility Note

The companion script `supplementary_ool_10.py` tests a minimal host–parasite competition gate with explicit retention, boundary specificity, and a two-dimensional spatial boundary diagnostic. It writes `parasitic_threshold_boundary_logic.png` under `outputs/paper_10/`, plus well-mixed and spatial CSV/JSON summaries. No notebook is required.

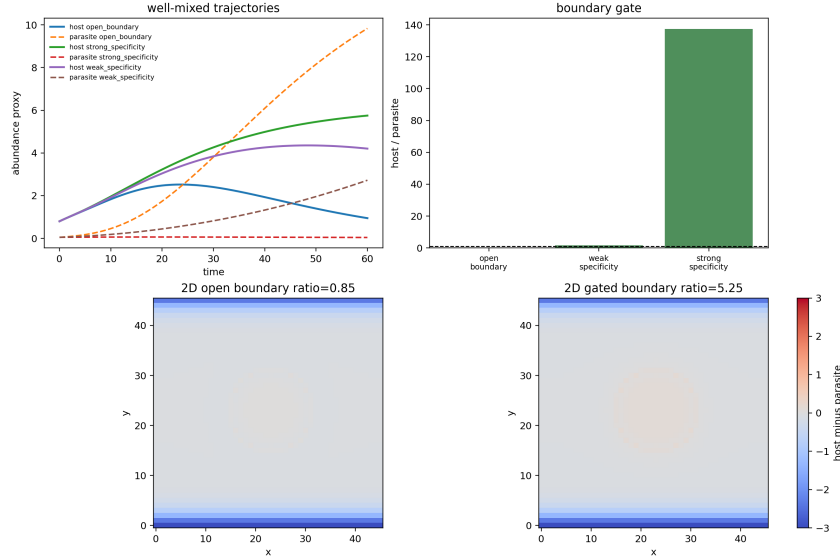


Figure 1: Parasitic-threshold diagnostic. The toy model compares host persistence with and without boundary specificity when a noncontributing replicator accesses the same resource stream, then shows a two-dimensional boundary-gate stress test. The figure is a mechanism diagnostic, not a claim about modern viruses.

Related literature

This manuscript is self-contained within the Part II origins-of-life sequence. Shared CDFD/CDFL symbols (Φ , C , S , M_s , Ψ_s) are used as bookkeeping only; they do not require Part I physics results. Field anchors are cited in the body and bibliography.

Competing Interests

The author declares no competing interests.

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Data and Code Availability

Companion scripts, generated figures, and numerical outputs for this manuscript are included in the public CDFD Part II release archive (<https://github.com/bampita-bico/CDFD-Part-II-Release>;

Zenodo DOI 10.5281/zenodo.20264779). See the Reproducibility Note above for the paper-local script path.

Author Contributions

Conceptualization, formal analysis, computational diagnostics, and manuscript preparation:
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